Лабораторная работа 8

Адресация IPv4 и IPv6. Настройка маршрутизации

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Цель работы

Цель

Изучение принципов маршрутизации в IPv4- и IPv6-сетях и принципов настройки сетевого оборудования.

Задачи

Основные задачи

1. Развертывание стенда в GNS3 с четырьмя маршрутизаторами FRR.
2. Документирование топологии (схемы L1, L3) и таблиц адресации.
3. Настройка двойного стека IPv4 и IPv6.
4. Конфигурация RIP/RIPng для динамической маршрутизации.
5. Конфигурация OSPF/OSPFv3 для динамической маршрутизации.
6. Проверка связности и анализ маршрутов.

Стенд

Топология сети

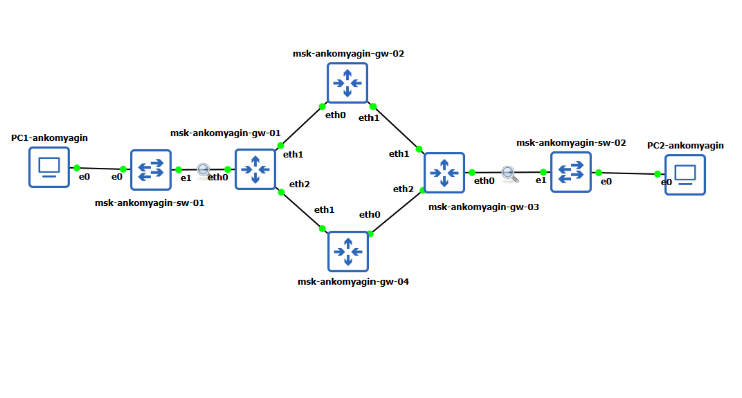


Рис. 1: Рис. 8.4: Топология сети из четырех маршрутизаторов в кольцевой конфигурации

Компоненты стенда

- **Маршрутизаторы FRR**: msk-ankomyagin-gw-01, msk-ankomyagin-gw-02, msk-ankomyagin-gw-03, msk-ankomyagin-gw-04;
- **Хосты VPCS**: PC1-ankomyagin, PC2-ankomyagin;
- **Коммутаторы**: msk-ankomyagin-sw-01, msk-ankomyagin-sw-02.

Схема подключения (L1)

PC1-ankomyagin - PuTTY

```
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 10.0.10.10/24 10.0.10.1
Checking for duplicate address...
VPCS : 10.0.10.10 255.255.255.0 gateway 10.0.10.1

VPCS> save
Saving startup configuration to startup.vpc
.  done

VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 10.0.10.10/24
GATEWAY     : 10.0.10.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 10018
RHOST:PORT  : 127.0.0.1:10019
MTU         : 1500
```

PC2-ankomyagin - PuTTY

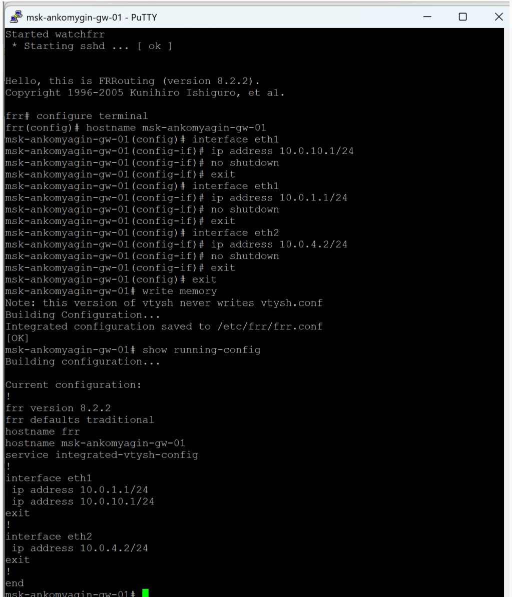
```
VPCS> ip 10.0.11.10/24 10.0.11.1
Checking for duplicate address...
VPCS : 10.0.11.10 255.255.255.0 gateway 10.0.11.1

VPCS> savr
Bad command: "savr". Use ? for help.

VPCS> save
Saving startup configuration to startup.vpc
.  done

VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 10.0.11.10/24
GATEWAY     : 10.0.11.1
DNS         :
MAC         : 00:50:79:66:68:01
LPORT       : 10016
RHOST:PORT  : 127.0.0.1:10017
MTU         : 1500
```

Адресация

Таблица сетей (табл. 8.1)

Сегмент	IPv4	IPv6
PC1 – gw-01	10.0.10.0/24	2001:10::/64
PC2 – gw-03	10.0.11.0/24	2001:11::/64
gw-01 – gw-02	10.0.1.0/24	2001:1::/64
gw-02 – gw-03	10.0.2.0/24	2001:2::/64
gw-03 – gw-04	10.0.3.0/24	2001:3::/64
gw-04 – gw-01	10.0.4.0/24	2001:4::/64

Схема сетевой адресации (L3)



msk-ankomygin-gw-01 - PuTTY

```
Started watchfrr
 * Starting sshd ... [ ok ]


Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# hostname msk-ankomyagin-gw-01
msk-ankomyagin-gw-01(config)# interface eth1
msk-ankomyagin-gw-01(config-if)# ip address 10.0.10.1/24
msk-ankomyagin-gw-01(config-if)# no shutdown
msk-ankomyagin-gw-01(config-if)# exit
msk-ankomyagin-gw-01(config)# interface eth1
msk-ankomyagin-gw-01(config-if)# ip address 10.0.1.1/24
msk-ankomyagin-gw-01(config-if)# no shutdown
msk-ankomyagin-gw-01(config-if)# exit
msk-ankomyagin-gw-01(config)# interface eth2
msk-ankomyagin-gw-01(config-if)# ip address 10.0.4.2/24
msk-ankomyagin-gw-01(config-if)# no shutdown
msk-ankomyagin-gw-01(config-if)# exit
msk-ankomyagin-gw-01(config)# exit
msk-ankomyagin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-ankomyagin-gw-01
service integrated-vtysh-config
!
interface eth1
 ip address 10.0.1.1/24
 ip address 10.0.10.1/24
exit
!
interface eth2
 ip address 10.0.4.2/24
exit
!
end
msk-ankomyagin-gw-01#
```

Построение стенда

Создание проекта в GNS3

msk-ankomyagin-gw-03 - PuTTY

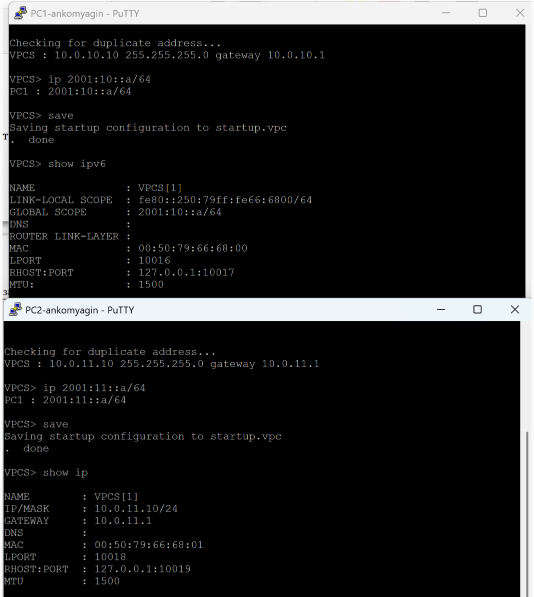
```
frr# configure terminal
frr(config)# hostname msk-ankomyagin-gw-03
msk-ankomyagin-gw-03(config)# interface eth0
msk-ankomyagin-gw-03(config-if)# ip address 10.0.11.1/24
msk-ankomyagin-gw-03(config-if)# no shutdown
msk-ankomyagin-gw-03(config-if)# exit
msk-ankomyagin-gw-03(config)# interface eth1
msk-ankomyagin-gw-03(config-if)# ip address 10.0.2.2/24
msk-ankomyagin-gw-03(config-if)# no shutdown
msk-ankomyagin-gw-03(config-if)# exit
msk-ankomyagin-gw-03(config)# interface eth2
msk-ankomyagin-gw-03(config-if)# ip address 10.0.3.1/24
msk-ankomyagin-gw-03(config-if)# no shutdown
msk-ankomyagin-gw-03(config-if)# exit
msk-ankomyagin-gw-03(config)# exit
msk-ankomyagin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
shoBuilding Configuration...
wIntegrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-ankomyagin-gw-03
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.11.1/24
exit
!
interface eth1
 ip address 10.0.2.2/24
exit
!
interface eth2
 ip address 10.0.3.1/24
exit
!
end
msk-ankomyagin-gw-03#
```

Переименование устройств

msk-ankomyagin-gw-04 - PuTTY

```
frr# configure terminal
frr(config)# hostname msk-ankomyagin-gw-04
msk-ankomyagin-gw-04(config)# interface eth0
msk-ankomyagin-gw-04(config-if)# ip address 10.0.3.2/24
msk-ankomyagin-gw-04(config-if)# no shutdown
msk-ankomyagin-gw-04(config-if)# exit
msk-ankomyagin-gw-04(config)# interface eth1
msk-ankomyagin-gw-04(config-if)# ip address 10.0.4.1/24
msk-ankomyagin-gw-04(config-if)# no shutdown
msk-ankomyagin-gw-04(config-if)# exit
msk-ankomyagin-gw-04(config)# write memory
% Unknown command: write memory
msk-ankomyagin-gw-04(config)# exit
msk-ankomyagin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-ankomyagin-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
exit
!
interface eth1
 ip address 10.0.4.1/24
exit
!
end
msk-ankomyagin-gw-04#
```

Включение захвата трафика



Конфигурация IPv4

Настройка на VPCS (PC1, PC2)

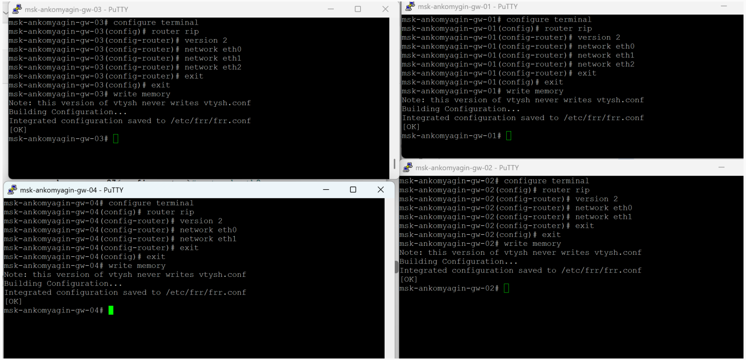
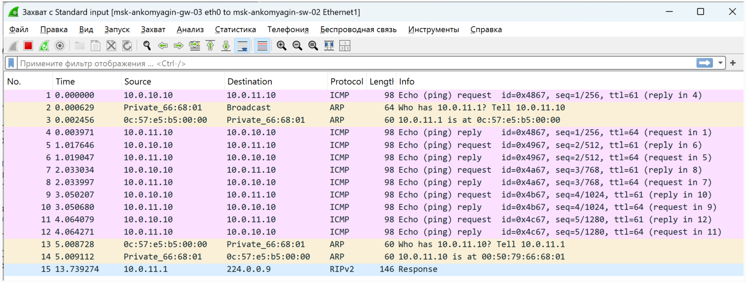


Рис. 7: Рис. 8.2: Присвоение IPv4-адресов оконечным устройствам: PC1 = 10.0.10.10/24, PC2 = 10.0.11.10/24



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x4867, seq=1/256, ttl=61 (reply in 4)
2	0.000629	Private_66:68:01	Broadcast	ARP	64	Who has 10.0.11.1? Tell 10.0.11.10
3	0.002456	0c:57:e5:b5:00:00	Private_66:68:01	ARP	60	10.0.11.1 is at 0c:57:e5:b5:00:00
4	0.003971	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x4867, seq=1/256, ttl=64 (request in 1)
5	1.017646	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x4967, seq=2/512, ttl=61 (reply in 6)
6	1.019047	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x4967, seq=2/512, ttl=64 (request in 5)
7	2.033034	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x4a67, seq=3/768, ttl=61 (reply in 8)
8	2.033997	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x4a67, seq=3/768, ttl=64 (request in 7)
9	3.050207	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x4b67, seq=4/1024, ttl=61 (reply in 10)
10	3.050680	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x4b67, seq=4/1024, ttl=64 (request in 9)
11	4.064079	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x4c67, seq=5/1280, ttl=61 (reply in 12)
12	4.064271	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x4c67, seq=5/1280, ttl=64 (request in 11)
13	5.008728	0c:57:e5:b5:00:00	Private_66:68:01	ARP	60	Who has 10.0.11.10? Tell 10.0.11.1
14	5.009112	Private_66:68:01	0c:57:e5:b5:00:00	ARP	60	10.0.11.10 is at 00:50:79:66:68:01
15	13.739274	10.0.11.1	224.0.0.9	RIPv2	146	Response

Рис. 8: Конфигурация IPv4-адресов на интерфейсах маршрутизаторов согласно табл. 8.2

Конфигурация IPv6

Включение маршрутизации и RA

Захват с Standard input [msk-ankomyagin-gw-03 eth0 to msk-ankomyagin-sw-02 Ethernet1]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения … <Ctrl-/>

No.	Time	Source	Destination	Protocol	Lengtl	Info
120	644.889312	10.0.11.1	224.0.0.9	RIPv2	106	Response
121	644.928994	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0xca69, seq=4, hop limit=61 (reply in 122)
122	644.929289	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0xca69, seq=4, hop limit=61 (request in 121)
123	645.948447	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0xca69, seq=5, hop limit=61 (reply in 124)
124	645.949001	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0xca69, seq=5, hop limit=61 (request in 123)
125	654.903159	10.0.11.1	224.0.0.9	RIPv2	146	Response
126	657.513268	fe80::e57:e5ff:feb5…	ff02::9	RIPng	86	Command Response, Version 1
127	658.917486	10.0.11.1	224.0.0.9	RIPv2	106	Response
128	661.487918	2001:10::a	2001:11::a	UDP	126	27792 → 27793 Len=64
129	661.488403	2001:11::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)[Malformed Packet]
130	661.503313	2001:10::a	2001:11::a	UDP	126	27792 → 27793 Len=64
131	661.503508	2001:11::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)[Malformed Packet]
132	661.522954	2001:10::a	2001:11::a	UDP	126	27792 → 27793 Len=64
133	661.524996	2001:11::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)[Malformed Packet]
134	666.526981	fe80::e57:e5ff:feb5…	2001:11::a	ICMPv6	86	Neighbor Solicitation for 2001:11::a from 0c:57:e5:b5:00:00
135	667.568915	fe80::e57:e5ff:feb5…	2001:11::a	ICMPv6	86	Neighbor Solicitation for 2001:11::a from 0c:57:e5:b5:00:00
136	668.607606	fe80::e57:e5ff:feb5…	2001:11::a	ICMPv6	86	Neighbor Solicitation for 2001:11::a from 0c:57:e5:b5:00:00
137	669.000436	fe80::e57:e5ff:feb5…	ff02::9	RIPng	186	Command Response, Version 1
138	679.911372	10.0.11.1	224.0.0.9	RIPv2	146	Response
139	700.008266	fe80::e57:e5ff:feb5…	ff02::9	RIPng	186	Command Response, Version 1
140	712.917352	10.0.11.1	224.0.0.9	RIPv2	146	Response
141	725.017226	fe80::e57:e5ff:feb5…	ff02::9	RIPng	186	Command Response, Version 1
142	737.740436	fe80::e57:e5ff:feb5…	ff02::1:ff00:a	ICMPv6	86	Neighbor Solicitation for 2001:11::a from 0c:57:e5:b5:00:00
143	737.742183	2001:11::a	fe80::e57:e5ff:feb5…	ICMPv6	86	Neighbor Advertisement 2001:11::a (sol, ovr) is at 00:50:79:66:68:01
144	737.747028	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x296a, seq=1, hop limit=61 (reply in 145)
145	737.747983	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x296a, seq=1, hop limit=61 (request in 144)
146	738.763907	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x296a, seq=2, hop limit=61 (reply in 147)
147	738.765250	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x296a, seq=2, hop limit=61 (request in 146)
148	739.783592	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x296a, seq=3, hop limit=61 (reply in 149)
149	739.784009	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x296a, seq=3, hop limit=61 (request in 148)
150	740.635221	10.0.11.1	224.0.0.9	RIPv2	146	Response
151	740.803883	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x296a, seq=4, hop limit=61 (reply in 152)
152	740.804272	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x296a, seq=4, hop limit=61 (request in 151)
153	741.822632	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x296a, seq=5, hop limit=61 (reply in 154)
154	741.824262	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x296a, seq=5, hop limit=61 (request in 153)

Рис. 9: Активация ipv6 forwarding и настройка Router Advertisement для автоматической раздачи

Назначение IPv6-адресов и префиксов

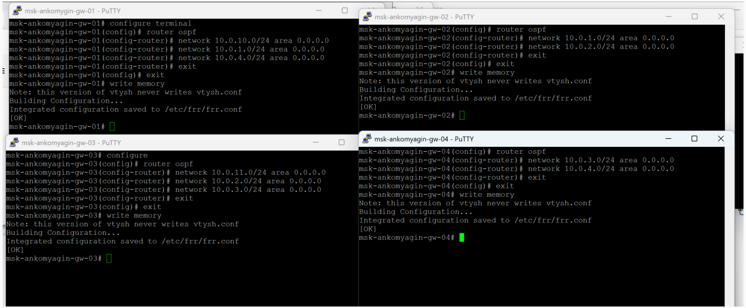
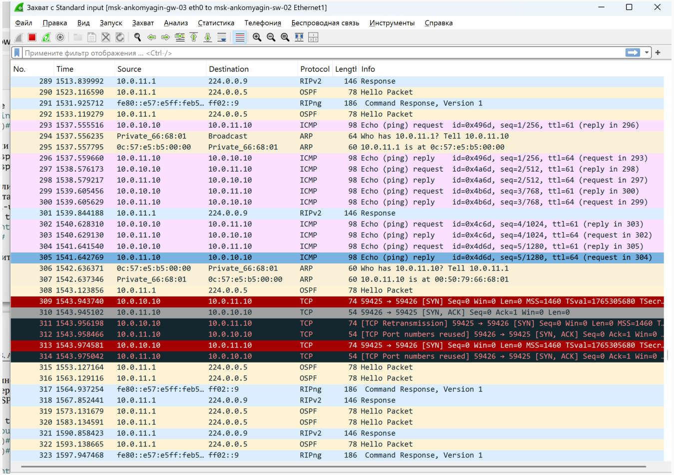


Рис. 10: Назначение IPv6-адресов на интерфейсы маршрутизаторов и настройка параметров ND

RIP: Динамическая маршрутизация IPv4



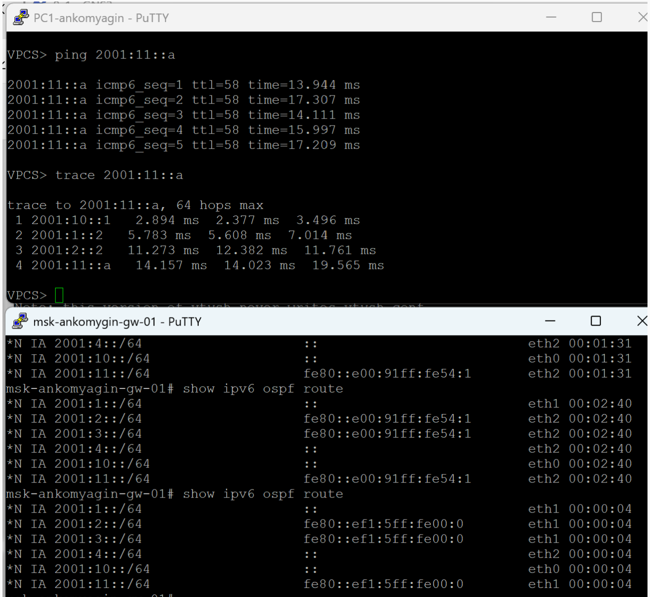
Захват с Standard input [msk-ankomyagin-gw-03 eth0 to msk-ankomyagin-sw-02 Ethernet1]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения … <Ctrl-/>

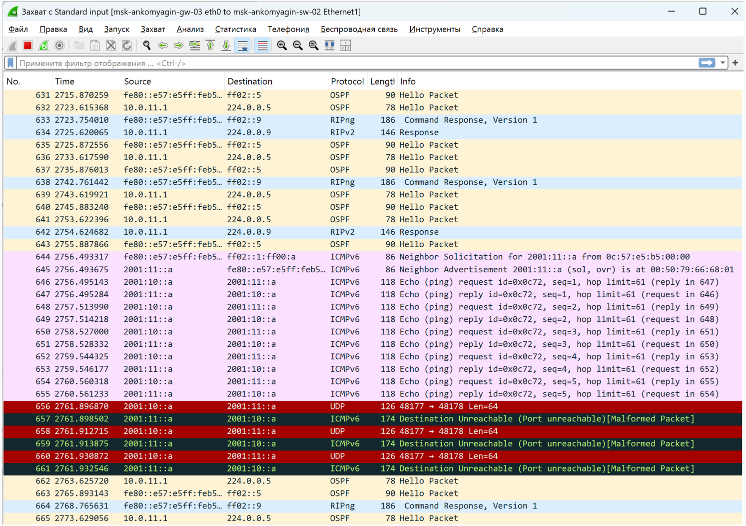
No.	Time	Source	Destination	Protocol	Lengtl	Info
289	1513.839992	10.0.11.1	224.0.0.9	RIPv2	146	Response
290	1523.116590	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
291	1531.925712	fe80::e57:e5ff:feb5…	ff02::9	RIPng	186	Command Response, Version 1
292	1533.119279	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
293	1537.555516	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x496d, seq=1/256, ttl=61 (reply in 296)
294	1537.556235	Private_66:68:01	Broadcast	ARP	64	Who has 10.0.11.1? Tell 10.0.11.10
295	1537.557795	0c:57:e5:b5:00:00	Private_66:68:01	ARP	60	10.0.11.1 is at 0c:57:e5:b5:00:00
296	1537.559660	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x496d, seq=1/256, ttl=64 (request in 293)
297	1538.576173	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x4a6d, seq=2/512, ttl=61 (reply in 298)
298	1538.579217	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x4a6d, seq=2/512, ttl=64 (request in 297)
299	1539.605456	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x4b6d, seq=3/768, ttl=61 (reply in 300)
300	1539.605629	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x4b6d, seq=3/768, ttl=64 (request in 299)
301	1539.844188	10.0.11.1	224.0.0.9	RIPv2	146	Response
302	1540.628310	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x4c6d, seq=4/1024, ttl=61 (reply in 303)
303	1540.629130	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x4c6d, seq=4/1024, ttl=64 (request in 302)
304	1541.641540	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x4d6d, seq=5/1280, ttl=61 (reply in 305)
305	1541.642769	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x4d6d, seq=5/1280, ttl=64 (request in 304)
306	1542.636371	0c:57:e5:b5:00:00	Private_66:68:01	ARP	60	Who has 10.0.11.10? Tell 10.0.11.1
307	1542.637346	Private_66:68:01	0c:57:e5:b5:00:00	ARP	60	10.0.11.10 is at 00:50:79:66:68:01
308	1543.123856	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
309	1543.943740	10.0.10.10	10.0.11.10	TCP	74	59425 → 59426 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1765305680 TSecr…
310	1543.945102	10.0.11.10	10.0.10.10	TCP	54	59426 → 59425 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
311	1543.956198	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 59425 → 59426 [SYN] Seq=0 Win=0 Len=0 MSS=1460 T…
312	1543.958466	10.0.11.10	10.0.10.10	TCP	54	[TCP Port numbers reused] 59426 → 59425 [SYN, ACK] Seq=0 Ack=1 Win=0 …
313	1543.974581	10.0.10.10	10.0.11.10	TCP	74	59425 → 59426 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1765305680 TSecr…
314	1543.975042	10.0.11.10	10.0.10.10	TCP	54	[TCP Port numbers reused] 59426 → 59425 [SYN, ACK] Seq=0 Ack=1 Win=0 …
315	1553.127164	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
316	1563.129116	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
317	1564.937254	fe80::e57:e5ff:feb5…	ff02::9	RIPng	186	Command Response, Version 1
318	1567.852441	10.0.11.1	224.0.0.9	RIPv2	146	Response
319	1573.131679	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
320	1583.134591	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
321	1590.858423	10.0.11.1	224.0.0.9	RIPv2	146	Response
322	1593.138665	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
323	1597.947468	fe80::e57:e5ff:feb5…	ff02::9	RIPng	186	Command Response, Version 1

Рис. 11: Рис. 8.8: Включение RIP версии 2 и объявление сетей через интерфейсы на маршрутизаторах


PC1-ankomyagin - PuTTY
VPCS> ping 2001:11::a
2001:11::a icmp6_seq=1 ttl=58 time=13.944 ms
2001:11::a icmp6_seq=2 ttl=58 time=17.307 ms
2001:11::a icmp6_seq=3 ttl=58 time=14.111 ms
2001:11::a icmp6_seq=4 ttl=58 time=15.997 ms
2001:11::a icmp6_seq=5 ttl=58 time=17.209 ms
VPCS> trace 2001:11::a
trace to 2001:11::a, 64 hops max
1 2001:10::1 2.894 ms 2.377 ms 3.496 ms
2 2001:1::2 5.783 ms 5.608 ms 7.014 ms
3 2001:2::2 11.273 ms 12.382 ms 11.761 ms
4 2001:11::a 14.157 ms 14.023 ms 19.565 ms
VPCS>
msk-ankomyagin-gw-01 - PuTTY
*N IA 2001:4::/64 :: eth2 00:01:31
*N IA 2001:10::/64 :: eth0 00:01:31
*N IA 2001:11::/64 fe80::e00:91ff:fe54:1 eth2 00:01:31
msk-ankomyagin-gw-01# show ipv6 ospf route
*N IA 2001:1::/64 :: eth1 00:02:40
*N IA 2001:2::/64 fe80::e00:91ff:fe54:1 eth2 00:02:40
*N IA 2001:3::/64 fe80::e00:91ff:fe54:1 eth2 00:02:40
*N IA 2001:4::/64 :: eth2 00:02:40
*N IA 2001:10::/64 :: eth0 00:02:40
*N IA 2001:11::/64 fe80::e00:91ff:fe54:1 eth2 00:02:40
msk-ankomyagin-gw-01# show ipv6 ospf route
*N IA 2001:1::/64 :: eth1 00:00:04
*N IA 2001:2::/64 fe80::ef1:5ff:fe00:0 eth1 00:00:04
*N IA 2001:3::/64 fe80::ef1:5ff:fe00:0 eth1 00:00:04
*N IA 2001:4::/64 :: eth2 00:00:04
*N IA 2001:10::/64 :: eth0 00:00:04
*N IA 2001:11::/64 fe80::ef1:5ff:fe00:0 eth1 00:00:04

RIP: Тестирование связности IPv4

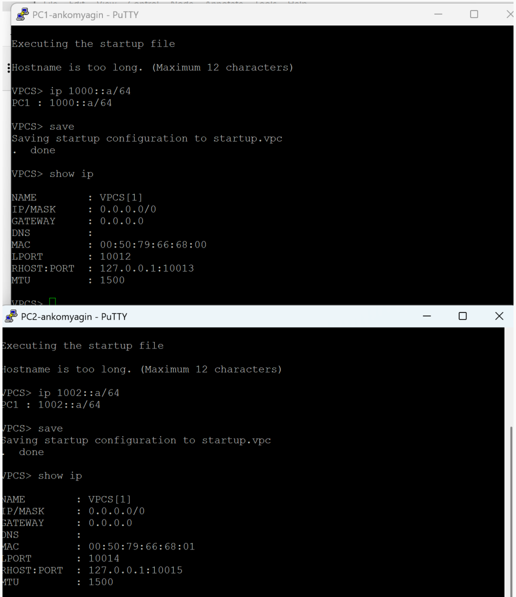
Ping и trace между PC1 и PC2



No.	Time	Source	Destination	Protocol	Lengtl	Info
631	2715.870259	fe80::e57:e5ff:feb5…	ff02::5	OSPF	90	Hello Packet
632	2723.615368	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
633	2723.754010	fe80::e57:e5ff:feb5…	ff02::9	RIPng	186	Command Response, Version 1
634	2725.620065	10.0.11.1	224.0.0.9	RIPv2	146	Response
635	2725.872556	fe80::e57:e5ff:feb5…	ff02::5	OSPF	90	Hello Packet
636	2733.617590	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
637	2735.876013	fe80::e57:e5ff:feb5…	ff02::5	OSPF	90	Hello Packet
638	2742.761442	fe80::e57:e5ff:feb5…	ff02::9	RIPng	186	Command Response, Version 1
639	2743.619921	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
640	2745.883240	fe80::e57:e5ff:feb5…	ff02::5	OSPF	90	Hello Packet
641	2753.622396	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
642	2754.624682	10.0.11.1	224.0.0.9	RIPv2	146	Response
643	2755.887866	fe80::e57:e5ff:feb5…	ff02::5	OSPF	90	Hello Packet
644	2756.493317	fe80::e57:e5ff:feb5…	ff02::1:ff00:a	ICMPv6	86	Neighbor Solicitation for 2001:11::a from 0c:57:e5:b5:00:00
645	2756.493675	2001:11::a	fe80::e57:e5ff:feb5…	ICMPv6	86	Neighbor Advertisement 2001:11::a (sol, ovr) is at 00:50:79:66:68:01
646	2756.495143	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x0c72, seq=1, hop limit=61 (reply in 647)
647	2756.495284	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x0c72, seq=1, hop limit=61 (request in 646)
648	2757.513990	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x0c72, seq=2, hop limit=61 (reply in 649)
649	2757.514218	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x0c72, seq=2, hop limit=61 (request in 648)
650	2758.527000	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x0c72, seq=3, hop limit=61 (reply in 651)
651	2758.528332	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x0c72, seq=3, hop limit=61 (request in 650)
652	2759.544325	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x0c72, seq=4, hop limit=61 (reply in 653)
653	2759.546177	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x0c72, seq=4, hop limit=61 (request in 652)
654	2760.560318	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x0c72, seq=5, hop limit=61 (reply in 655)
655	2760.561233	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x0c72, seq=5, hop limit=61 (request in 654)
656	2761.896870	2001:10::a	2001:11::a	UDP	126	48177 → 48178 Len=64
657	2761.898502	2001:11::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)[Malformed Packet]
658	2761.912715	2001:10::a	2001:11::a	UDP	126	48177 → 48178 Len=64
659	2761.913875	2001:11::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)[Malformed Packet]
660	2761.930872	2001:10::a	2001:11::a	UDP	126	48177 → 48178 Len=64
661	2761.932546	2001:11::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)[Malformed Packet]
662	2763.625720	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
663	2765.893143	fe80::e57:e5ff:feb5…	ff02::5	OSPF	90	Hello Packet
664	2768.765631	fe80::e57:e5ff:feb5…	ff02::9	RIPng	186	Command Response, Version 1
665	2773.629056	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet

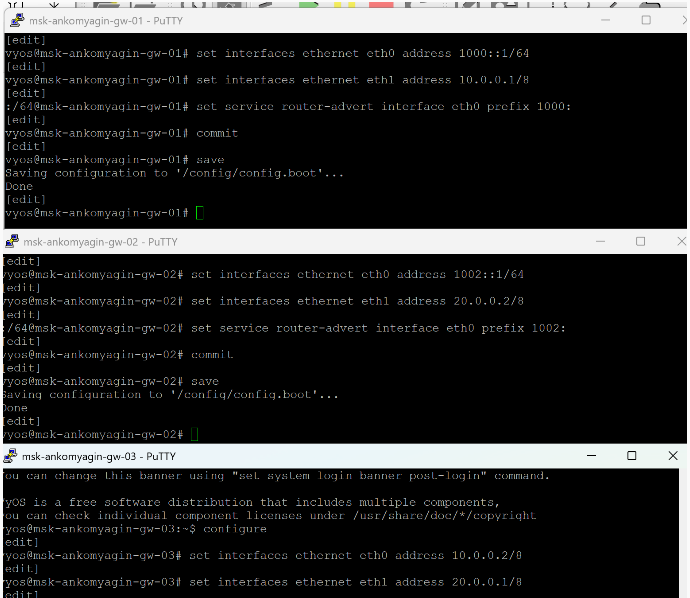
Рис. 13: Проверка пути маршрутизации: PC1 пингует PC2 (10.0.11.10), trace определяет переходы через маршрутизаторы

Проверка отказоустойчивости

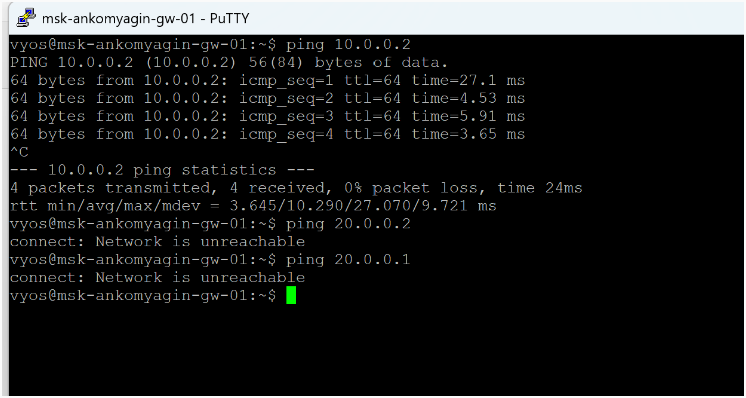


RIP: Анализ трафика IPv4

Захват RIPv2-обновлений



Наблюдение за сходимостью сети




```
vyos@msk-ankomyagin-gw-01:~$ ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=27.1 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=4.53 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=5.91 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=3.65 ms
^C
--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 24ms
rtt min/avg/max/mdev = 3.645/10.290/27.070/9.721 ms
vyos@msk-ankomyagin-gw-01:~$ ping 20.0.0.2
connect: Network is unreachable
vyos@msk-ankomyagin-gw-01:~$ ping 20.0.0.1
connect: Network is unreachable
vyos@msk-ankomyagin-gw-01:~$
```

Рис. 16: Изменение объявлений маршрутов при отказе интерфейса и его восстановлении

RIPng: Динамическая маршрутизация IPv6

Конфигурация RIPng

msk-ankomyagin-gw-01 - PuTTY

```
vyos@msk-ankomyagin-gw-01:~$ configure
[edit]
vyos@msk-ankomyagin-gw-01# set protocols rip network 10.0.0.0/8
[edit]
vyos@msk-ankomyagin-gw-01# commit
[edit]
vyos@msk-ankomyagin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
```

msk-ankomyagin-gw-02 - PuTTY

```
[edit]
vyos@msk-ankomyagin-gw-02# set protocols rip network 20.0.0.0/8
[edit]
vyos@msk-ankomyagin-gw-02# commit
[edit]
vyos@msk-ankomyagin-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-ankomyagin-gw-02#
```

msk-ankomyagin-gw-03 - PuTTY

```
vyos@msk-ankomyagin-gw-03:~$ configure
[edit]
vyos@msk-ankomyagin-gw-03# set protocols rip network 10.0.0.0/8
[edit]
vyos@msk-ankomyagin-gw-03# set protocols rip network 20.0.0.0/8
[edit]
vyos@msk-ankomyagin-gw-03# commit
```

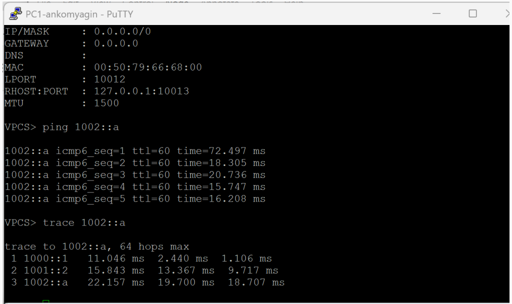
Проверка маршрутов RIPng

```
vyos@msk-ankomyagin-gw-01# ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=18.9 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=10.6 ms
^C
--- 10.0.0.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 8ms
rtt min/avg/max/mdev = 10.575/14.723/18.872/4.150 ms
[edit]
vyos@msk-ankomyagin-gw-01# ping 20.0.0.1
PING 20.0.0.1 (20.0.0.1) 56(84) bytes of data.
64 bytes from 20.0.0.1: icmp_seq=1 ttl=64 time=4.92 ms
64 bytes from 20.0.0.1: icmp_seq=2 ttl=64 time=6.16 ms
64 bytes from 20.0.0.1: icmp_seq=3 ttl=64 time=3.27 ms
^C
--- 20.0.0.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 12ms
rtt min/avg/max/mdev = 3.266/4.780/6.155/1.186 ms
[edit]
vyos@msk-ankomyagin-gw-01# ping 20.0.0.2
PING 20.0.0.2 (20.0.0.2) 56(84) bytes of data.
64 bytes from 20.0.0.2: icmp_seq=1 ttl=63 time=19.8 ms
64 bytes from 20.0.0.2: icmp_seq=2 ttl=63 time=6.69 ms
64 bytes from 20.0.0.2: icmp_seq=3 ttl=63 time=11.9 ms
64 bytes from 20.0.0.2: icmp_seq=4 ttl=63 time=8.53 ms
^C
--- 20.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 19ms
rtt min/avg/max/mdev = 6.685/11.708/19.752/5.001 ms
[edit]
vyos@msk-ankomyagin-gw-01#
```

Рис. 18: Вывод команды show ipv6 ripng для проверки таблиц маршрутизации IPv6

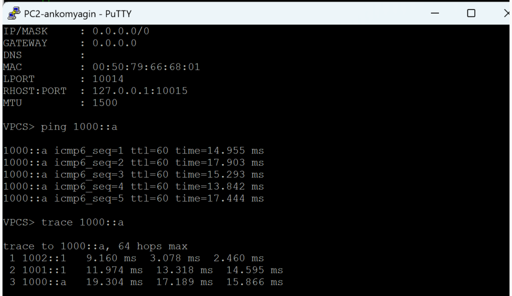
RIPng: Тестирование связности IPv6

Ping и trace между PC1 и PC2 (IPv6)



PC1-ankomyagin - PuTTY

```
IP/MASK      : 0.0.0.0/0
GATEWAY      : 0.0.0.0
DNS          :
MAC          : 00:50:79:66:68:00
LPORT        : 10012
RHOST:PORT   : 127.0.0.1:10013
MTU          : 1500

VPCS> ping 1002::a

1002::a icmp6_seq=1 ttl=60 time=72.497 ms
1002::a icmp6_seq=2 ttl=60 time=18.305 ms
1002::a icmp6_seq=3 ttl=60 time=20.736 ms
1002::a icmp6_seq=4 ttl=60 time=15.747 ms
1002::a icmp6_seq=5 ttl=60 time=16.208 ms

VPCS> trace 1002::a

trace to 1002::a, 64 hops max
 1 1000::1    11.046 ms  2.440 ms  1.106 ms
 2 1001::2    15.843 ms  13.367 ms  9.717 ms
 3 1002::a    22.157 ms  19.700 ms  18.707 ms
```



PC2-ankomyagin - PuTTY

```
IP/MASK      : 0.0.0.0/0
GATEWAY      : 0.0.0.0
DNS          :
MAC          : 00:50:79:66:68:01
LPORT        : 10014
RHOST:PORT   : 127.0.0.1:10015
MTU          : 1500

VPCS> ping 1000::a

1000::a icmp6_seq=1 ttl=60 time=14.955 ms
1000::a icmp6_seq=2 ttl=60 time=17.903 ms
1000::a icmp6_seq=3 ttl=60 time=15.293 ms
1000::a icmp6_seq=4 ttl=60 time=13.842 ms
1000::a icmp6_seq=5 ttl=60 time=17.444 ms

VPCS> trace 1000::a

trace to 1000::a, 64 hops max
 1 1002::1    9.160 ms  3.078 ms  2.460 ms
 2 1001::1    11.974 ms  13.318 ms  14.595 ms
 3 1000::a    19.304 ms  17.189 ms  15.866 ms
```

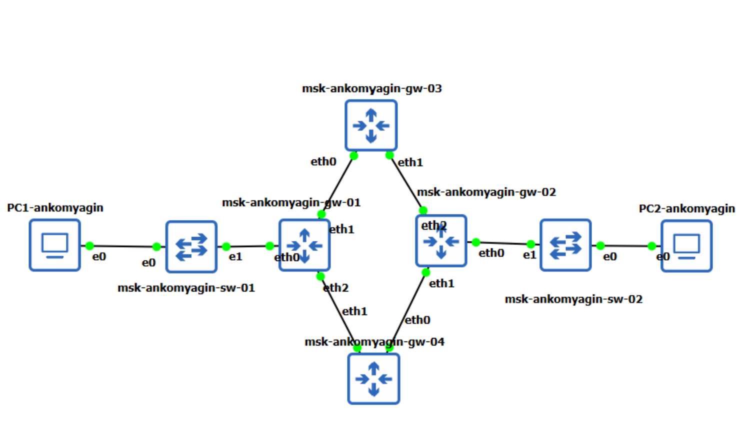


Рис. 20: Отключение интерфейса eth0 на маршрутизаторе (RIPng) и восстановление IPv6-маршрута

RIPng: Анализ трафика IPv6

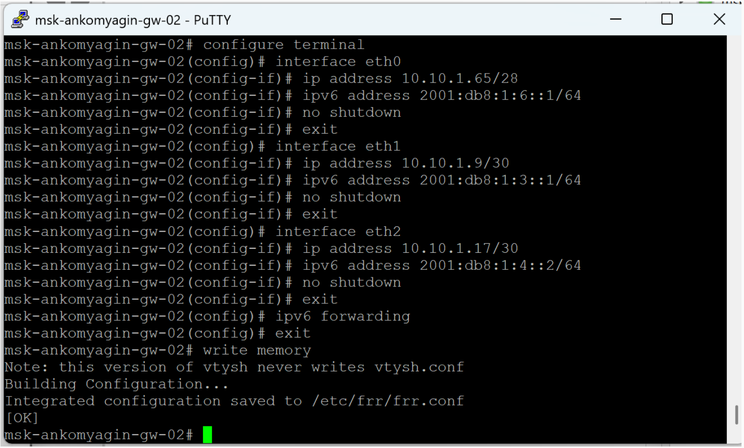
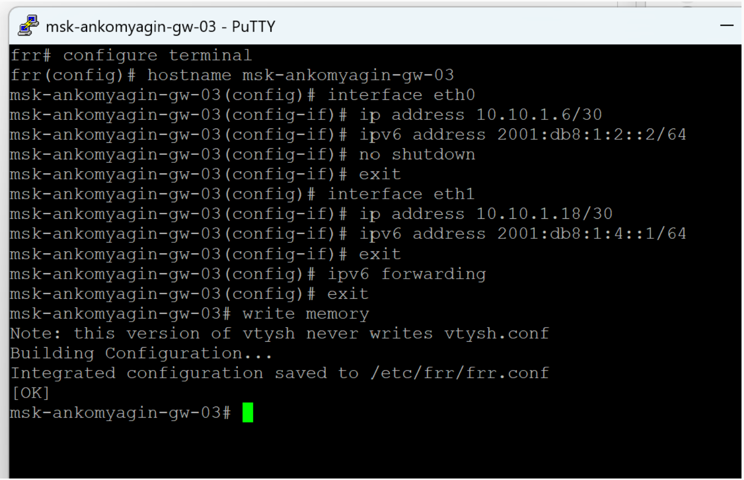


Рис. 21: Multicast FF02::9: IPv6-обновления маршрутизации RIPng и обмены информацией о сетях

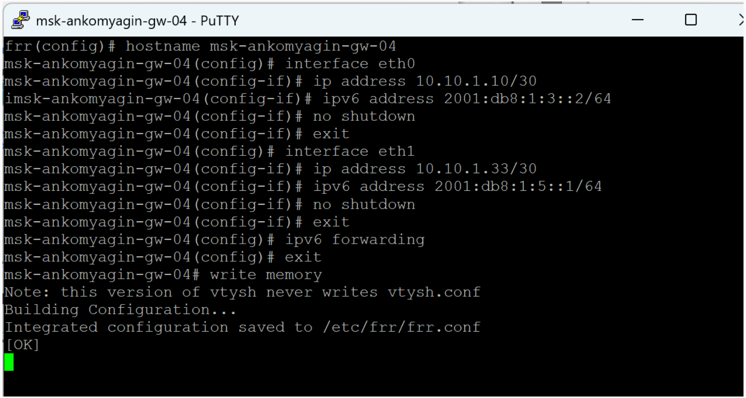
Служебный трафик при сходимости



```
msk-ankomyagin-gw-03 - PuTTY
frr# configure terminal
frr(config)# hostname msk-ankomyagin-gw-03
msk-ankomyagin-gw-03(config)# interface eth0
msk-ankomyagin-gw-03(config-if)# ip address 10.10.1.6/30
msk-ankomyagin-gw-03(config-if)# ipv6 address 2001:db8:1:2::2/64
msk-ankomyagin-gw-03(config-if)# no shutdown
msk-ankomyagin-gw-03(config-if)# exit
msk-ankomyagin-gw-03(config)# interface eth1
msk-ankomyagin-gw-03(config-if)# ip address 10.10.1.18/30
msk-ankomyagin-gw-03(config-if)# ipv6 address 2001:db8:1:4::1/64
msk-ankomyagin-gw-03(config-if)# exit
msk-ankomyagin-gw-03(config)# ipv6 forwarding
msk-ankomyagin-gw-03(config)# exit
msk-ankomyagin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-03#
```

Рис. 22: ICMPv6 и NDP-запросы при перестроении IPv6-маршрутов после отказа

OSPF: Динамическая маршрутизация IPv4



```
frr(config)# hostname msk-ankomyagin-gw-04
msk-ankomyagin-gw-04(config)# interface eth0
msk-ankomyagin-gw-04(config-if)# ip address 10.10.1.10/30
imsk-ankomyagin-gw-04(config-if)# ipv6 address 2001:db8:1:3::2/64
msk-ankomyagin-gw-04(config-if)# no shutdown
msk-ankomyagin-gw-04(config-if)# exit
msk-ankomyagin-gw-04(config)# interface eth1
msk-ankomyagin-gw-04(config-if)# ip address 10.10.1.33/30
msk-ankomyagin-gw-04(config-if)# ipv6 address 2001:db8:1:5::1/64
msk-ankomyagin-gw-04(config-if)# no shutdown
msk-ankomyagin-gw-04(config-if)# exit
msk-ankomyagin-gw-04(config)# ipv6 forwarding
msk-ankomyagin-gw-04(config)# exit
msk-ankomyagin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
```

Рис. 23: Включение OSPF (OSPFv2) и объявление сетей в area 0.0.0.0 на маршрутизаторах

Проверка соседства и маршрутов OSPF

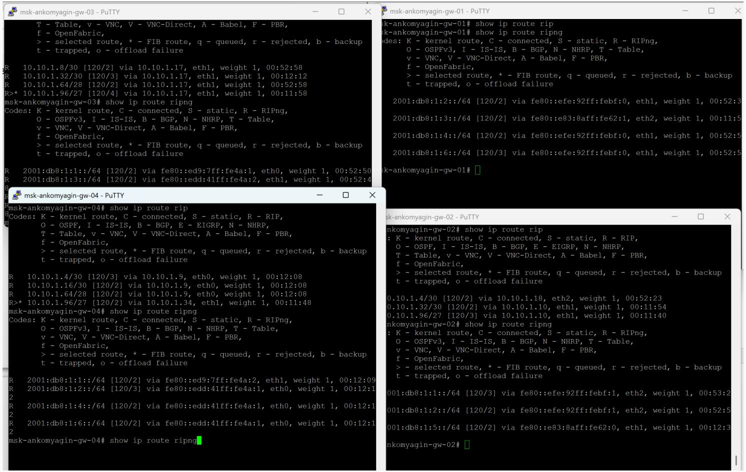


Рис. 24: Вывод команд show ip ospf neighbor и show ip ospf route для анализа соседей и маршрутов

OSPF: Тестирование связности IPv4

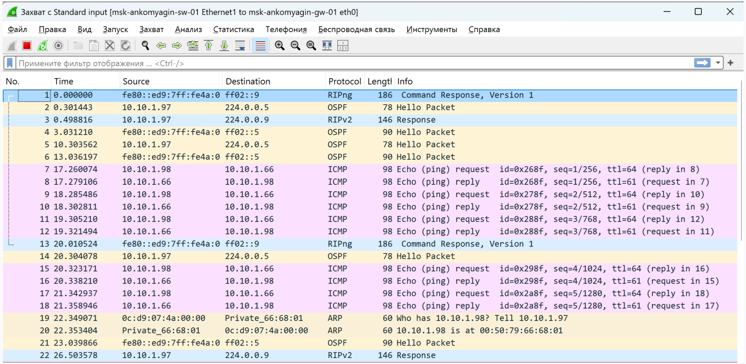
Ping и trace маршрутов OSPF

PC1-ankomyagin - PuTTY

```
84 bytes from 10.10.1.66 icmp_seq=5 ttl=61 time=12.061 ms

VPCS> ping 10.10.1.66

84 bytes from 10.10.1.66 icmp_seq=1 ttl=61 time=25.504 ms
84 bytes from 10.10.1.66 icmp_seq=2 ttl=61 time=17.964 ms
84 bytes from 10.10.1.66 icmp_seq=3 ttl=61 time=17.738 ms
84 bytes from 10.10.1.66 icmp_seq=4 ttl=61 time=15.597 ms
84 bytes from 10.10.1.66 icmp_seq=5 ttl=61 time=17.847 ms

VPCS> trace 10.10.1.66
trace to 10.10.1.66, 8 hops max, press Ctrl+C to stop
 1   10.10.1.97    4.839 ms  2.716 ms  4.778 ms
 2   10.10.1.6    10.073 ms  6.736 ms  10.537 ms
```

Рис. 25: Проверка маршрутов и альтернативных путей при использовании OSPF (IPv4, 224.0.0.5/224.0.0.6)

Анализ служебных обменов OSPF



Захват с Standard input [msk-ankomyagin-sw-01 Ethernet1 to msk-ankomyagin-gw-01 eth0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Lengtl	Info
1	0.000000	fe80::ed9:7ff:fe4a:0	ff02::9	RIPng	186	Command Response, Version 1
2	0.301443	10.10.1.97	224.0.0.5	OSPF	78	Hello Packet
3	0.498816	10.10.1.97	224.0.0.9	RIPv2	146	Response
4	3.031210	fe80::ed9:7ff:fe4a:0	ff02::5	OSPF	90	Hello Packet
5	10.303562	10.10.1.97	224.0.0.5	OSPF	78	Hello Packet
6	13.036197	fe80::ed9:7ff:fe4a:0	ff02::5	OSPF	90	Hello Packet
7	17.260074	10.10.1.98	10.10.1.66	ICMP	98	Echo (ping) request id=0x268f, seq=1/256, ttl=64 (reply in 8)
8	17.279106	10.10.1.66	10.10.1.98	ICMP	98	Echo (ping) reply id=0x268f, seq=1/256, ttl=61 (request in 7)
9	18.285486	10.10.1.98	10.10.1.66	ICMP	98	Echo (ping) request id=0x278f, seq=2/512, ttl=64 (reply in 10)
10	18.302811	10.10.1.66	10.10.1.98	ICMP	98	Echo (ping) reply id=0x278f, seq=2/512, ttl=61 (request in 9)
11	19.305210	10.10.1.98	10.10.1.66	ICMP	98	Echo (ping) request id=0x288f, seq=3/768, ttl=64 (reply in 12)
12	19.321494	10.10.1.66	10.10.1.98	ICMP	98	Echo (ping) reply id=0x288f, seq=3/768, ttl=61 (request in 11)
13	20.010524	fe80::ed9:7ff:fe4a:0	ff02::9	RIPng	186	Command Response, Version 1
14	20.304078	10.10.1.97	224.0.0.5	OSPF	78	Hello Packet
15	20.323171	10.10.1.98	10.10.1.66	ICMP	98	Echo (ping) request id=0x298f, seq=4/1024, ttl=64 (reply in 16)
16	20.338210	10.10.1.66	10.10.1.98	ICMP	98	Echo (ping) reply id=0x298f, seq=4/1024, ttl=61 (request in 15)
17	21.342937	10.10.1.98	10.10.1.66	ICMP	98	Echo (ping) request id=0x2a8f, seq=5/1280, ttl=64 (reply in 18)
18	21.358946	10.10.1.66	10.10.1.98	ICMP	98	Echo (ping) reply id=0x2a8f, seq=5/1280, ttl=61 (request in 17)
19	22.349071	0c:d9:07:4a:00:00	Private_66:68:01	ARP	60	Who has 10.10.1.98? Tell 10.10.1.97
20	22.353404	Private_66:68:01	0c:d9:07:4a:00:00	ARP	60	10.10.1.98 is at 00:50:79:66:68:01
21	23.039866	fe80::ed9:7ff:fe4a:0	ff02::5	OSPF	90	Hello Packet
22	26.503578	10.10.1.97	224.0.0.9	RIPv2	146	Response

Рис. 26: Захват OSPF Hello, LSA и служебных пакетов при отказе интерфейса и восстановлении

OSPFv3: Динамическая маршрутизация IPv6

Конфигурация OSPFv3

PC1-ankomyagin - PuTTY

```
VPCS> ip 10.10.1.98/27 10.10.1.97
Checking for duplicate address...
VPCS : 10.10.1.98 255.255.255.224 gateway 10.10.1.97

VPCS> ip 2001:db8:1:1::a/64
PC1 : 2001:db8:1:1::a/64

VPCS> save
Saving startup configuration to startup.vpc
.  done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 2001:db8:1:1::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:01
LPORT               : 10096
RHOST:PORT          : 127.0.0.1:10097
MTU:                : 1500
```

PC2-ankomyagin - PuTTY

```
VPCS> ip 10.10.1.66/28 10.10.1.65
Checking for duplicate address...
VPCS : 10.10.1.66 255.255.255.240 gateway 10.10.1.65

VPCS> ip 2001:db8:1:6::a/64
PC1 : 2001:db8:1:6::a/64

VPCS> save
Saving startup configuration to startup.vpc
.  done

VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 10.10.1.66/28
GATEWAY     : 10.10.1.65
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 10098
RHOST:PORT  : 127.0.0.1:10099
MTU         : 1500
```

Проверка соседства IPv6

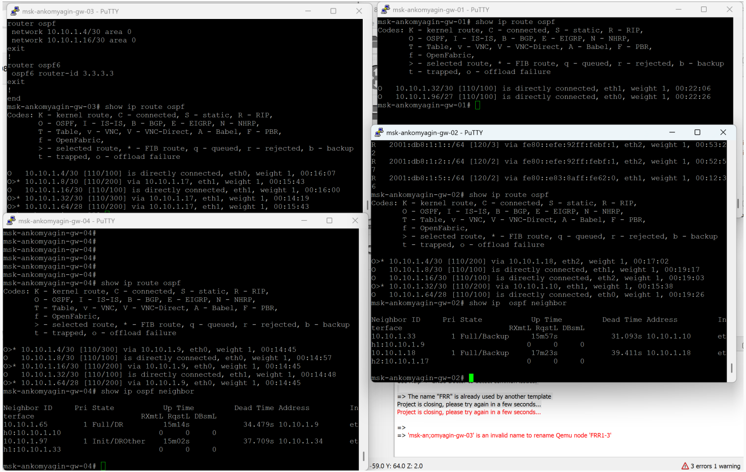


Рис. 28: Вывод команд show ipv6 ospf6 neighbor и show ipv6 ospf6 route для проверки соседей в IPv6

Результаты

Результаты

- Документирована топология: схемы L1, L3, таблицы адресации;
- Настроен двойной стек IPv4/IPv6 на четырёх маршрутизаторах;
- Реализована динамическая маршрутизация RIP/RIPng и OSPF/OSPFv3;
- Проверена связность между хостами с использованием ping/trace;
- Проведён анализ трафика и маршрутизации при отказах интерфейсов.

Выводы

Выводы

- **RIP/RIPv2** — простой дистанционно-векторный протокол (метрика = хопы), медленная сходимость (~180 сек);
- **RIPng** — расширение RIP для IPv6 (multicast FF02::9);
- **OSPF/OSPFv2** — протокол состояния канала с быстрой сходимостью, метрика = стоимость;
- **OSPFv3** — протокол состояния канала для IPv6 (multicast FF02::5/FF02::6);
- **Двойной стек IPv4/IPv6** — одновременная работа обоих протоколов в одной сети;
- **Анализ трафика** — служебные multicast-рассылки обеспечивают синхронизацию маршрутов и позволяют отследить сходимость после отказов.